Private 5G-Enabled Industrial AI Assistant Using Large Language Models and Retrieval-Augmented Generation

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Abstract:

The digital transformation of modern industries has led to the generation of extensive amounts of technical documentation, including manuals, procedures, maintenance guides, equipment specifications, and operational records. While this information is critical for the proper functioning and maintenance of industrial facilities, it can be challenging to navigate, requiring engineers and technicians to spend significant amounts of time searching for the required information. Current approaches to document search are limited in their ability to understand natural language questions and provide contextually relevant answers. Recent advancements in Large Language Models (LLMs) and Retrieval Augmented Generation enable the creation of an Intelligent Industrial Assistant that can leverage the information contained in technical documents to answer natural language questions while being hosted on a secure Private 5G network to ensure data privacy.

The proposed solution will be built using the Private 5G infrastructure and leverage Large Language Models to provide an answer to technical questions asked by the user. The initial step will involve preparing the set of equipment manuals, procedures, and reports that will be used to train the model. A vector database will be populated with information extracted from the provided documents, encoded as vectors using semantic embeddings. The user will be able to ask questions using a web interface, which will allow the system to query the vector database to identify related documents and use an LLM to generate contextually relevant answers. The application will be deployed as a set of REST API endpoints and will include additional features such as document preprocessing, semantic search, and a conversational interface. The project can be extended to include support for additional languages and voice commands and connected to equipment monitoring systems to provide real-time equipment insights.

The completed project will produce an Industrial AI Assistant that is capable of fulfilling conversational queries, retrieving relevant information from technical documents, and utilizing Large Language Models to reason about the information contained in them. The project will introduce the students to concepts of Generative AI while also providing opportunities for exploring the solutions in industrial settings.